NAME

frico − Friendly Radiation Integral COde

SYNOPSIS

frico [*OPTION*]...

DESCRIPTION

A 3D Method of Moments code for the simulation of antennas and scattering structures.

This code implements the Method of Moments (MoM) as described in "NASA Contractor Report 189594" titled "MOM3D Method of Moments Code - Theory Manual" by John F. Shaeffer.

The code takes as input a GMSH .geo file and some parameters specifying the sources and the simulation frequency. The output is written to text files (for impedance and radiation patterns) and SILO files for the visualization of fields using VisIt.

OPTIONS

−A Do not use quadratures, use approximate matrix as described in the MOM3D theory manual.

−d Dump matrices and right hand sides of all the linear systems.

−e Force reorientation of basis functions involved in the delta-gap (see CAVEATS).

−g <path>
Path to the input geometry as .geo file obtained from GMSH.

−f <frequency>
Simulation frequency specified in Hz. Exponential notation may be used. Mutually exclusive with -R.

−s <physgroup>
Currently only antenna simulation is supported, therefore only delta-gap sources are allowed. Delta-gap voltage is applied to one or more curves that need to be grouped in a physical group of your GMSH model. For example, if a delta-gap source on curves 4,5,6 and 7 is wanted, you need to put in your GMSH model something like

```
Physical Curve("source", 10) = {4,5,6,7};
```

−R <start>:<step>:<end>
Do a sweep from <start> Hz to <end> Hz at steps of <step> Hz. Frequencies may be specified using exponential notation. Mutually exclusive with -f.

−k <order>
Integration order to be used when not in approximate mode (-A). Default is 1.

−n <name>
Set a descriptive name for the simulation. Default is "default".

−S Expert stuff: In the current implementation the matrix is slightly nonsymmetric (only for non-approximate variant) because of the handling of the self elements. This flag forces symmetrization by doing 0.5*(Z + Z^T). Do not use.

−x <list of tags>
List of comma-separated tags corresponding to GMSH surfaces you want to exclude from loading. This is useful when in your mesh you have edges with more than two triangles attached, but the option exists only for convenience. The proper way to do this is to delete surfaces directly in the .geo file, for example:

```
SetFactory("OpenCASCADE");
Cylinder(1) = {0, 0, 0, 1, 0, 0, 0.01, 2*Pi};
Cylinder(2) = {-1, 0, 0, 1, 0, 0, 0.01, 2*Pi};
Coherence;
```

```
MeshSize{ PointsOf{ Surface{:}; } } = 0.01;
Mesh 2;
Delete { Volume{1,2}; }
Delete { Surface{3}; }
```

−Z <value>
Set system impedance for SWR calculations (default: 50 Ohm)

CAVEATS

Delta-gap sources can occasionally give some problems. What happens is the following:

- The delta-gap is applied on a curve between two entities with different tags

  In this case it is assumed (this should be confirmed with GMSH folks) that all the triangles in the lower-numbered entity have lower tags than the triangles in the higher-numbered entity. In this case there is no problem, because all the RWG basis functions are oriented lower-to-higher and therefore all the edges in the delta-gap will have coherent orientation.

- The delta-gap is applied on curves embedded into a single entity **AND** the curves do not form a closed curve.

  In this case the triangles on both sides of the curves will be part of the same entity, therefore it is not possible to guarantee a coherent orientation of the involved RWG basis functions. In turn, this makes impossible to know the sign of the voltage to apply to the edges of the delta-gap. The code employs a simple algorithm that checks the orientation of adjacent basis functions, and reorients them if it is incoherent. The algorithm works in most cases, but its robustness w.r.t. real-world geometries has not been thoroughly tested.

- Fancier cases

  Not explicitly supported for now, therefore YMMV. Please file a bug describing your situation.

Be aware of this issue if you get nonsensical results when using delta-gaps. This situation can be easily detected by eye by using the "Label" plot in VisIt to inspect the numbering of the elements across the delta-gap.

SEE ALSO

The required theory to understand this code and MoM in general is contained in

- "Field Computations by Moment Methods" by Roger F. Harrington.
- "NASA Contractor Report 189594" titled "MOM3D Method of Moments Code - Theory Manual" by John F. Shaeffer.
- "The Method of Moments in Electromagnetics" by K4LLA Walton C. Gibson.

You should visit the Friuli Venezia Giulia region in Italy and taste "Frico", which is one of the main regional dishes.

BUGS

Report bugs to <matteo.cicuttin@polito.it>

AUTHOR

Written by IV3IWE Matteo Cicuttin

COPYRIGHT